Thm: (Karras, Kreck, Neumann, Ossa)
80's "SK book"

The only SK invariants of closed n-dim oriented manifolds are

- χ Euler char
- σ signature

<u>Def</u>: $\mathcal{M}_n$ = monoid of diffeo classes of compact oriented n-dim manifolds, $\sqcup$

$$SK_n := \frac{Gr(\mathcal{M}_n)}{SK \text{ equiv.}}$$

$\rightsquigarrow$ fully computed in SK gp

Scissors congruence spectra – brief history

Zakharevich: defines a SC K-theory spectrum via the notion of "assembler"

Grothendieck site + extra data

$\rightsquigarrow$ recovers on π_0 classical SC gps. for polyhedra + $K_0(Var) = \dfrac{\text{free ab gp on var}}{[X]+[Y \setminus X]=[Y]}$

Campbell

defines "subtractive cats" and K-theory spectra $\rightsquigarrow$ recovers $K(\mathrm{Var})$

$\rightsquigarrow$ their $K(\mathrm{Var})$ agree

Our goal:

can we construct a K-theory spectrum of n-dim manifolds w/ $\pi_0 = SK_n$?

Issue #1:

pieces in a cut & paste need to be objects in your category

$\Rightarrow$ need to work in $\mathcal{M}_n^\partial$

Solution to issue # 1:

Def: for $M, N \in \mathcal{M}_n^\partial$, define cut & paste relation the same way, s.t.

- do not allow boundaries to be cut
- all cut boundaries are required to be pasted back together

$$SK_n^\partial := Gr(\mathcal{M}_n^\partial) \big/ SK \text{ equiv.}$$

Rmk. different than the SK_n^∂ proposed in SK-book

$M_1 \cup_\emptyset M_2 \simeq_{SK} M_1 \amalg M_2$ ⚠ we do not allow this

Thm (Hoekzema, M, Murray, Rovi, Semikina)

$\exists$ split SES

$$\begin{array}{ccccccccc} & & & & [M] & \longrightarrow & [\partial M] & & \\ 0 & \to & SK_n & \xrightarrow{\alpha} & SK_n^{\partial} & \xrightarrow{\beta} & \underbrace{C_{n-1}} & \to & 0 \\ & & [M] & \longrightarrow & [M] & & & & \end{array}$$

Gr (monoid of diffeo classes of null cob $(n-1)$-dim manifolds)

Pf idea $\boxed{\ker\beta \subseteq \operatorname{im}\alpha}$

$\in \ker\beta$

$\|$ claim

Pf of claim: